Pulse Amplitude Modulation and Constellation

EE 471C / EE 381K-17 Wireless Communication Lab

From Previous Lecture

- It is possible to filter bandlimited continuous-time signals by operating directly on their samples
- The power spectrum is used to compute the spectrum of WSS random processes
- Different measures are bandwidth are used to define the practical bandwidth of a signal
- All wireless communication signals are passband
- Complex envelope is a convenient mathematical representation of passband signals at baseband

Preview of today's lecture

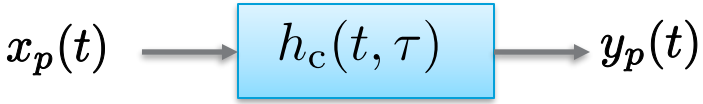
- Baseband equivalent channel
  - Use the complex envelope to represent a passband signal at baseband
- Linear complex pulse amplitude modulation techniques
  - Explain the concept of linear complex pulse amplitude modulation and relevant notation
- Constellations
  - Draw a constellation diagram including bit labels
  - Calculate the energy of a constellation and normalize it

Baseband equivalent channel

Learning objective:

- Use the complex envelope to represent a passband signal at baseband.

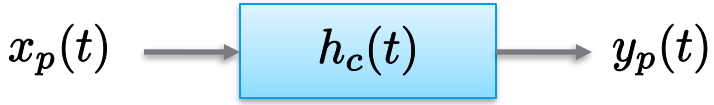
Linear time varying channel model



$$y_p = \int h_c(t,\tau) x_p(t-\tau) d\tau$$

- Linearity
  - Comes from the propagation medium
- Time varying
  - Mobility of the transmitter, receiver, or reflections
- Good for simulations, challenging to build algorithms or analysis
  - Reason is that the propagation channel must be estimated
  - Used in underwater communication, also proposed by Cohere for 5G

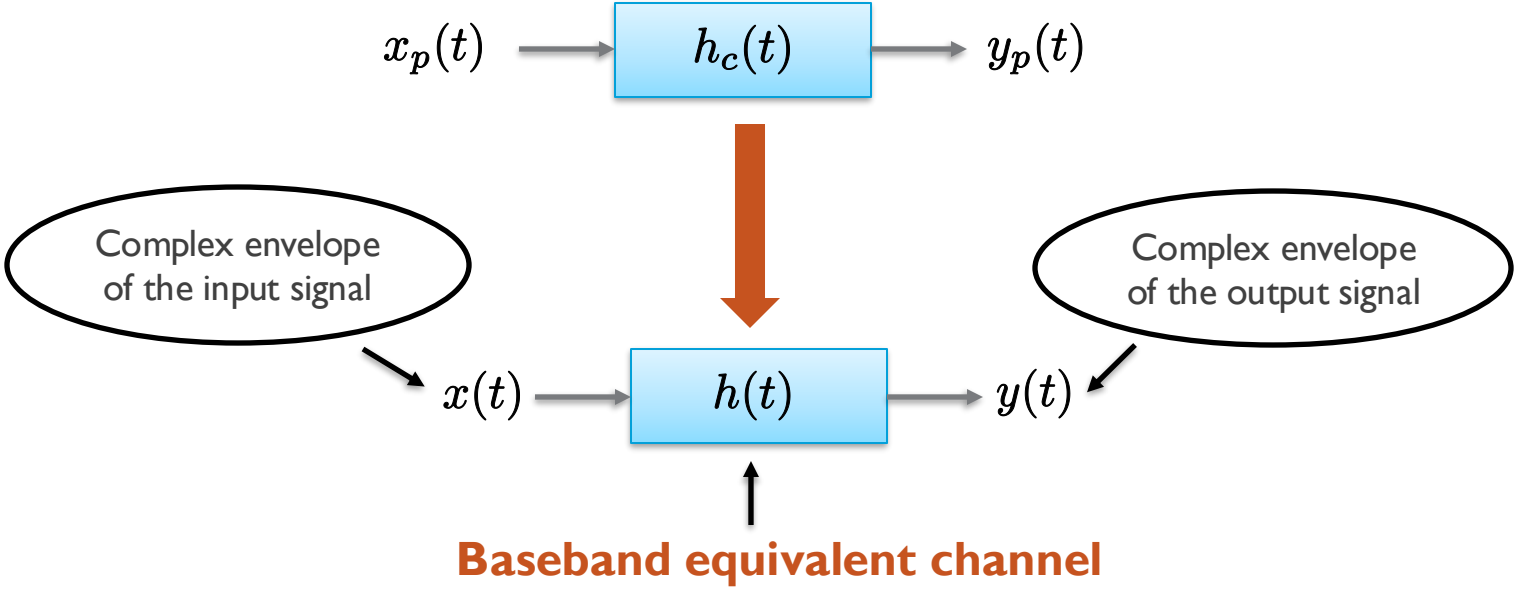
Linear time invariant channel



$$y_p(t) = \int h_c(t-\tau)x_p(\tau)d\tau$$

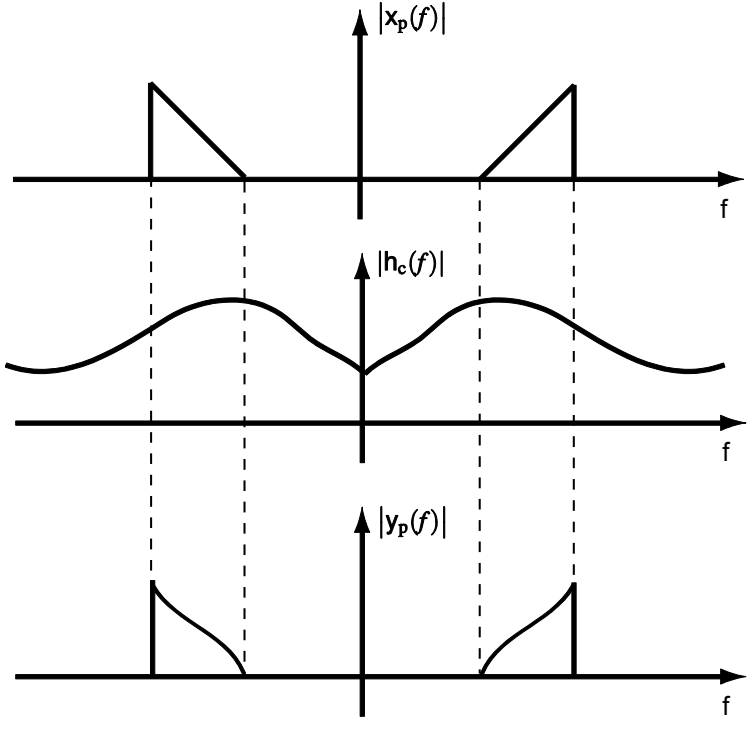
- Wireless systems are designed to enable a time-invariant assumption
  - Often called the *block fading* assumption
  - Applies for a certain number of packets, blocks, or frames
- Typical sizes are from 0.5ms to 10ms
  - Depends on the channel coherence time (quantified later in the course)

Baseband equivalent channel



Passband model for the wireless channel

Passband transmitted signal

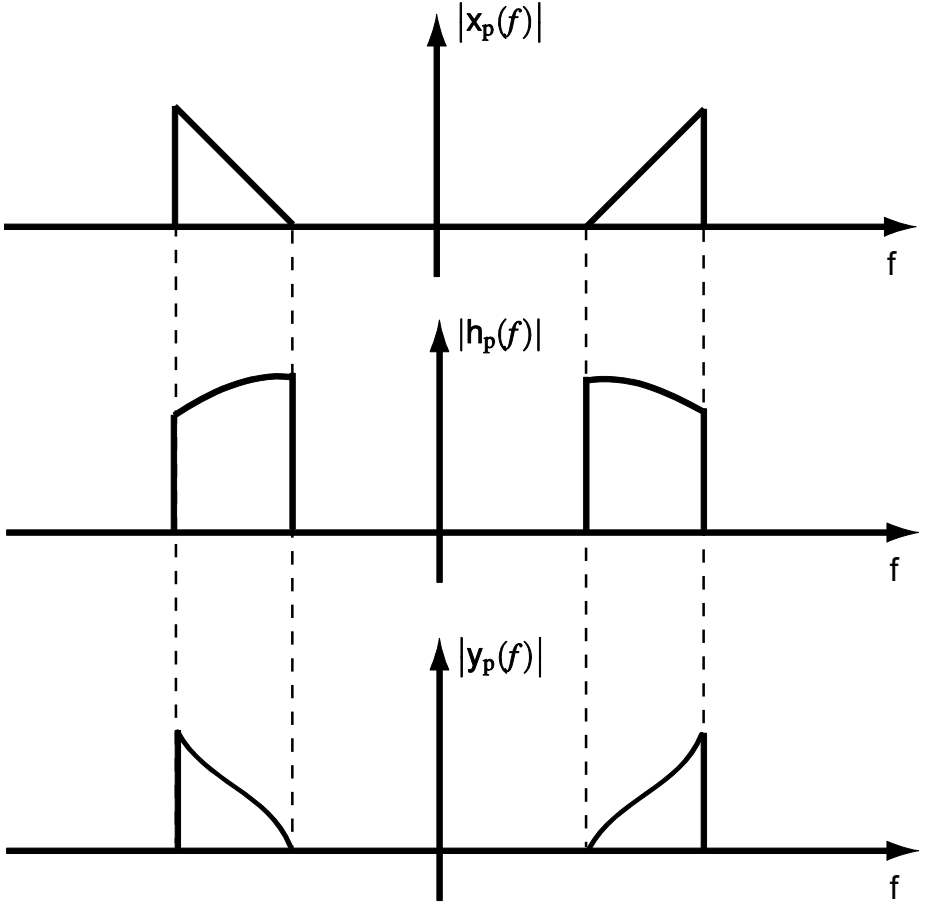


Passband received signal

$$y_p(t) = \int h_c(t-\tau)x_p(\tau)d\tau$$

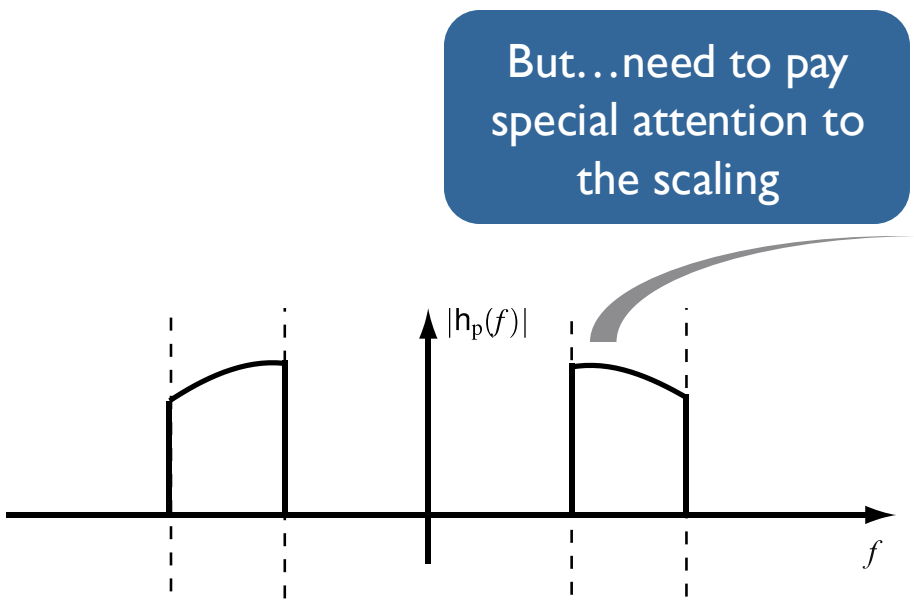
Only the passband channel portion is important

Equivalent passband channel

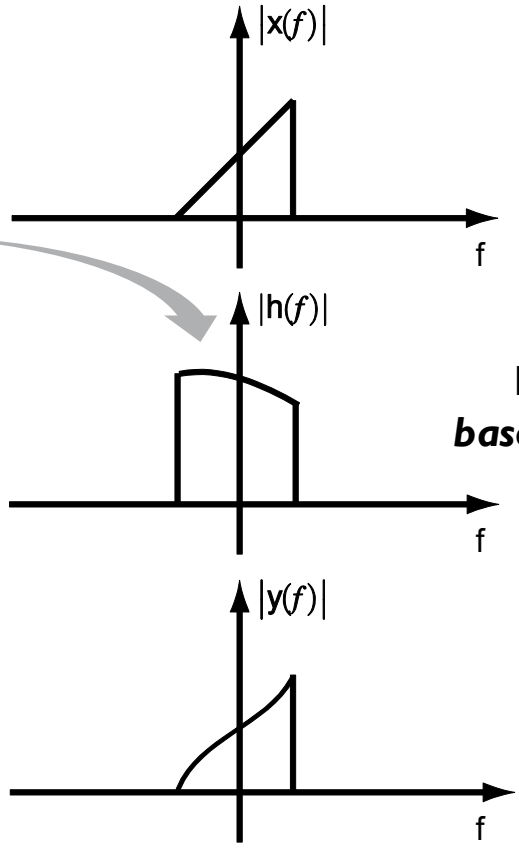


Use passband to find the baseband equivalent

But…need to pay special attention to the scaling

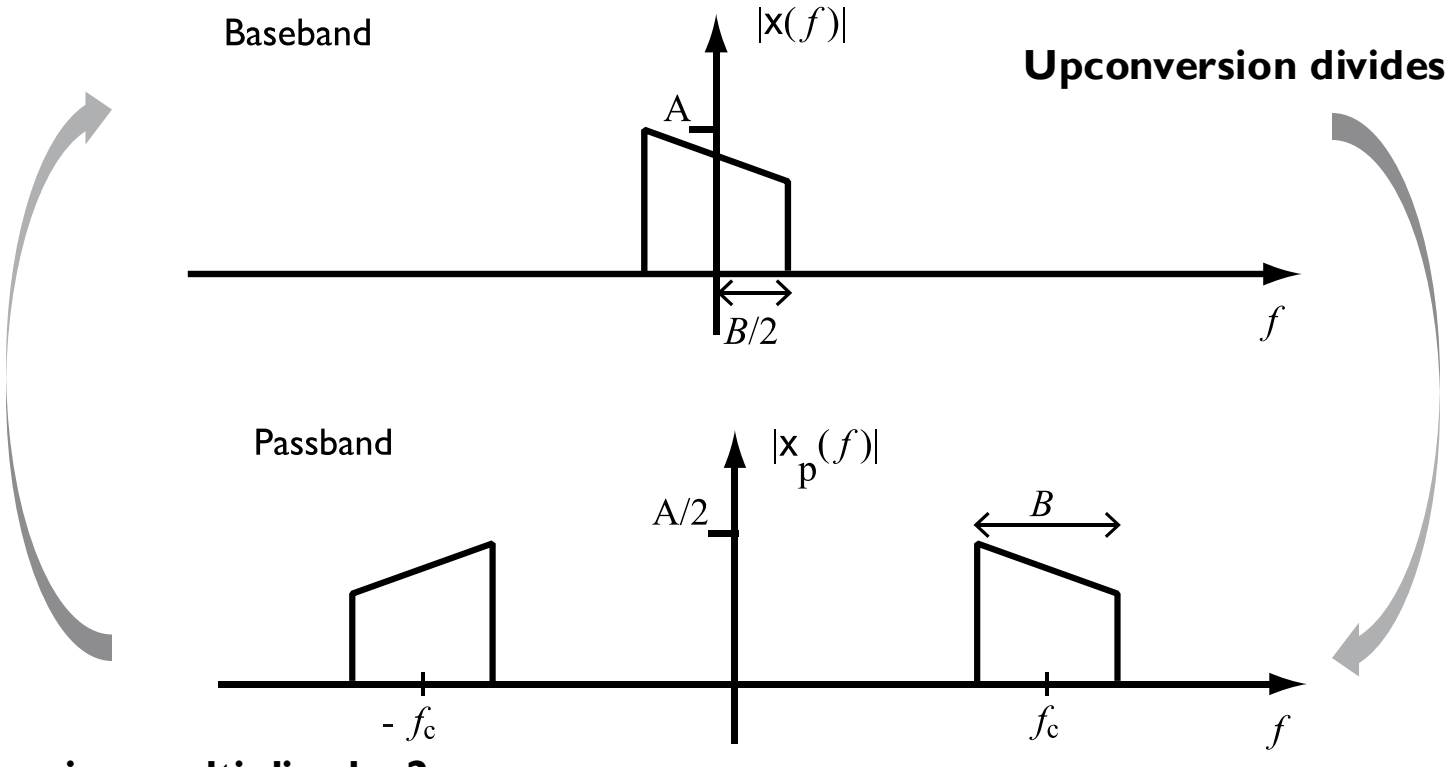


Equivalent *passband* channel



Equivalent *baseband* channel

Remembering the scaling factors for the signal



Slightly different result for the channel

Passband filtered channel

$$\mathsf{y}_\mathrm{p}(f) = \mathsf{h}_\mathrm{c}(f)\mathsf{x}_\mathrm{p}(f)$$

$$\begin{aligned}\mathsf{p}(f) &= \mathrm{rect}\left(\frac{f-f_\mathrm{c}}{B}\right) + \mathrm{rect}\left(\frac{-(f+f_\mathrm{c})}{B}\right)\\ &= \mathrm{rect}\left(\frac{f-f_\mathrm{c}}{B}\right) + \mathrm{rect}\left(\frac{f+f_\mathrm{c}}{B}\right)\end{aligned}$$

$$\begin{aligned}\mathsf{y}_\mathrm{p}(f) &= \mathsf{h}_\mathrm{c}(f)\mathsf{p}(f)\mathsf{x}_\mathrm{p}(f) \quad \Leftarrow \quad \mathsf{x}_\mathrm{p}(f) = \mathsf{x}_\mathrm{p}(f)\mathsf{p}(f)\\ &= \mathsf{h}_\mathrm{p}(f)\mathsf{x}_\mathrm{p}(f)\end{aligned}$$

$$\mathsf{h}_\mathrm{p}(f) = \mathsf{p}(f)\mathsf{h}_\mathrm{c}(f)$$

Figuring out the scaling factor

- From the perspective of converting from $y(f) = h(f)x(f)$

$$y_p(f) = \frac{1}{2}h(f - f_c)x(f - f_c) + \frac{1}{2}h^*(-f - f_c)x^*(-f - f_c)$$

- From the perspective of $y_p(f) = h_p(f)x_p(f)$

$$y_p(f) = \left(\frac{1}{2}h_b(f - f_c) + \frac{1}{2}h_b{}^*(-f - f_c)\right)\left(\frac{1}{2}x(f - f_c) + \frac{1}{2}x^*(-f - f_c)\right)$$

$$= \frac{1}{4}h_b(f - f_c)x(f - f_c) + \frac{1}{4}h_b{}^*(-f - f_c)x^*(-f - f_c).$$

$$\boxed{h(f) = \frac{1}{2}h_b(f)}$$

Baseband equivalent channel from the channel

Since $\mathsf{h}(f) = \frac{1}{2}\mathsf{h}_{\mathrm{b}}(f)$ where $\mathsf{h}_{\mathrm{b}}(f)$ is the baseband equivalent of $\mathsf{h}_{\mathrm{p}}(f)$

$$\mathsf{h}(f) = \frac{1}{2}\, 2\,\mathrm{rect}\left(\frac{f}{B}\right)\mathsf{h}_{\mathrm{p}}(f+f_{\mathrm{c}})$$

$$= \mathrm{rect}\left(\frac{f}{B}\right)\left[\mathrm{rect}\left(\frac{f}{B}\right)\mathsf{h}_{\mathrm{c}}(f+f_{\mathrm{c}}) + \mathrm{rect}\left(\frac{f+f_{\mathrm{c}}+f_{\mathrm{c}}}{B}\right)\mathsf{h}_{\mathrm{c}}(f-f_{\mathrm{c}})\right]$$

$$= \mathrm{rect}\left(\frac{f}{B}\right)\mathsf{h}_{\mathrm{c}}(f+f_{\mathrm{c}}).$$

Key time frequency domain result

Baseband equivalent channel in the time domain

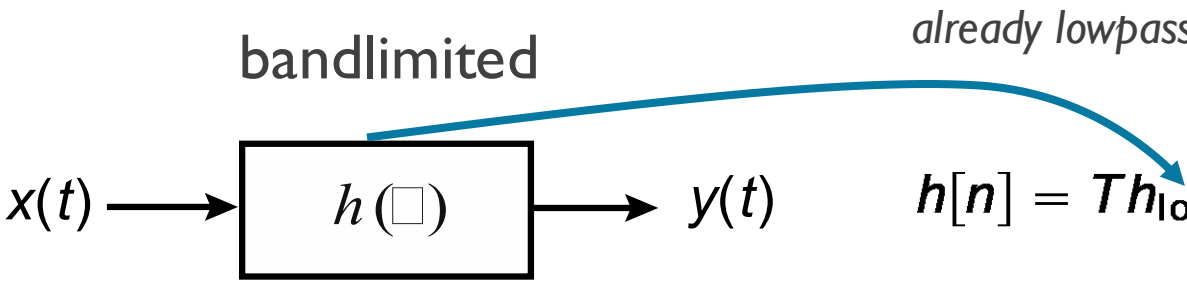
- In the frequency domain

$$\mathsf{h}(f) = \mathrm{rect}\left(\frac{f}{B}\right)\mathsf{h}_{\mathrm{c}}(f + f_{\mathrm{c}})$$

- In the time domain

$$\begin{aligned} h(t) &= B\mathrm{sinc}(Bt) * h_{\mathrm{c}}(t)e^{\mathrm{j}2\pi f_{\mathrm{c}}t} \\ &= B\int \mathrm{sinc}(B(t-\tau))h_{\mathrm{c}}(\tau)e^{\mathrm{j}2\pi f_{\mathrm{c}}\tau}\mathrm{d}\tau. \end{aligned}$$

Deriving the equivalent channel in discrete-time



Example

- Suppose that a signal has to propagate over a distance of $100m$ to arrive at the receiver and undergoes an attenuation of 0.1. The baseband signal bandwidth is $5MHz$ and the carrier frequency is $f_c = 2GHz$.
  - Find a linear time invariant model $h_c(t)$ for this channel
  - Find the passband channel $h_p(t)$
  - Find the baseband equivalent channel $h(t)$
  - Find the discrete-time equivalent channel $h[n]$

Example – finding the LTI model

This models the physical propagation delay and attenuation:

$$h_c(t) = \alpha\delta(t - \tau)$$

Where:

- $\alpha = 0.1$ (attenuation)
- $\tau = \frac{100}{c} \approx 3.33 \times 10^{-7}$ seconds (propagation delay, assuming speed of light $c = 3 \times 10^8$ m/s)

Baseband equivalent channel

Learning objective:

- Use the complex envelope to represent a passband signal at baseband.

Linear complex pulse amplitude modulation techniques

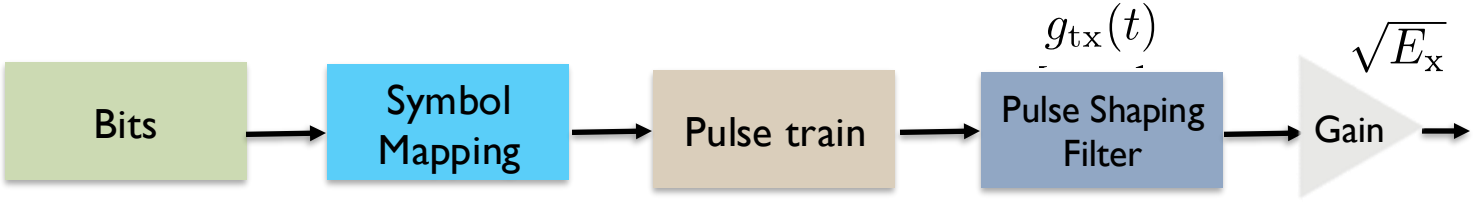
Learning objective:

- Explain the concept of linear complex pulse amplitude modulation and relevant notation.

Linear complex pulse amplitude modulation

- Digital modulation technique generated from linear combinations of shifted amplitude modulated pulses
- Include several practical linear modulation schemes of interest
  - QAM - quadrature amplitude modulation used in IEEE 802.11g, IEEE 802.16
  - PSK - phase shift keying used in EDGE
  - QPSK - quadrature phase shift keying (same as 4-QAM)
  - PAM - pulse amplitude modulation, used in wireline systems

Block diagram description



- Complex symbol at discrete time n: $s[n]$
- Symbol mapping takes a group of b bits and maps to a point in a constellation

  $\mathcal{C} = \{c_0, ..., c_{M-2}, c_{M-1}\} \qquad |\mathcal{C}| = M \qquad M = 2^b$

  - Symbol period: T
  - Symbol rate: $1/T$
  - Bit rate: b/T
- Complex equivalent baseband signal: $x(t) = \sqrt{E_{\mathrm{x}}} \sum_{n=-\infty}^{\infty} s[n] g_{\mathrm{tx}}(t - nT)$

Linear complex pulse amplitude modulation techniques

Learning objective:

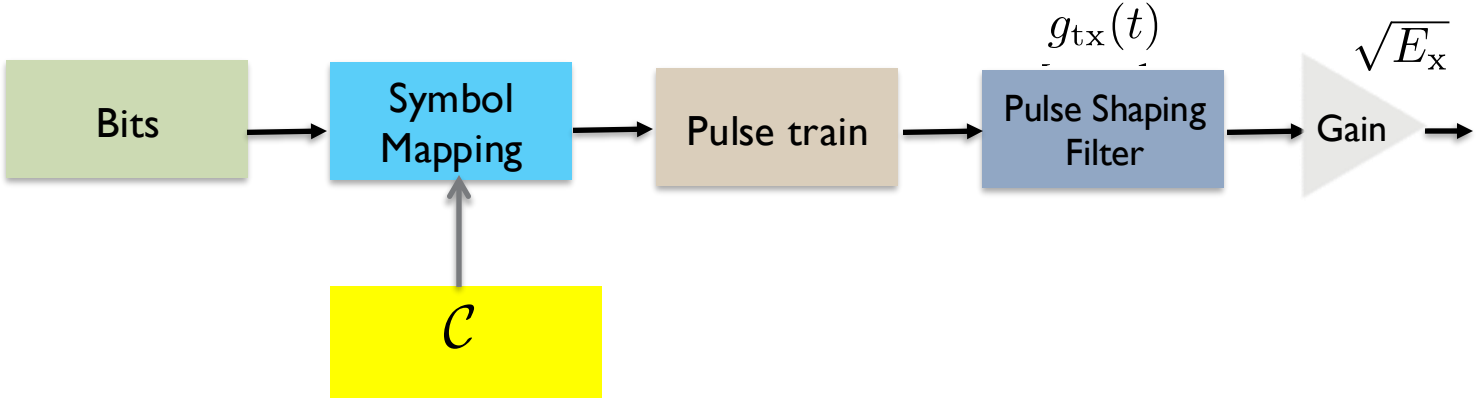
- Explain the concept of linear complex pulse amplitude modulation and relevant notation.

Constellations

Learning objectives:

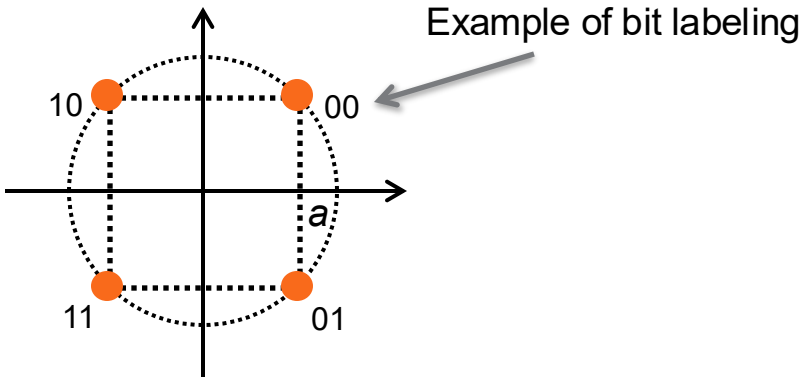
- Draw a constellation diagram including bit labels
- Calculate the energy of a constellation and normalize it

Complex pulse amplitude modulation



- Focus in this part on the constellation

Constellations



- A constellation is defined by
  - A set of symbols or alphabet
  - A bit-to-symbol mapping
- Bit labeling is typically done according to the Gray coding principle
  - Adjacent symbols differ by a single binary digit
- Assumptions made about constellations used in this course
  - They are equally likely
  - They are zero-mean
  - They are unit energy

Common constellations (1)

- Binary Phase Shift Keying (BPSK)

$$\mathcal{C} = \{+1, -1\}$$

- M-PAM is a generalization of BPSK to any M that is a power of 2

$$\mathcal{C} = \left\{ -\frac{(M-1)}{2}, \ldots, -\frac{1}{2}, \frac{1}{2}, \ldots, \frac{(M-1)}{2} \right\}$$ (not normalized)

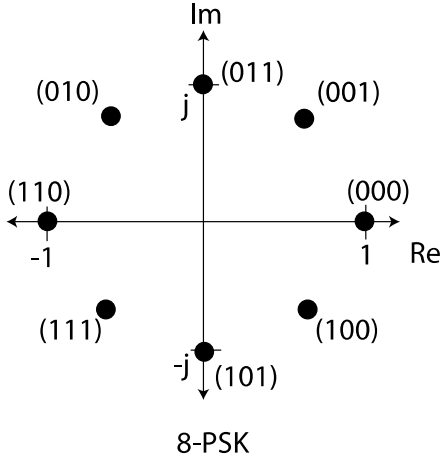
- Quadrature phase shift keying (QPSK) is a complex generalization of BPSK

$$\mathcal{C} = \{1 + \mathrm{j}, -1 + \mathrm{j}, -1 - \mathrm{j}, 1 - \mathrm{j}, \}$$ (not normalized)

Common constellations (2)

- M-Phase Shift Keying (M-PSK)

$$\mathcal{C} = \left\{ e^{\frac{\mathrm{j}2\pi k}{M}} \right\}_{k=0}^{M-1}$$



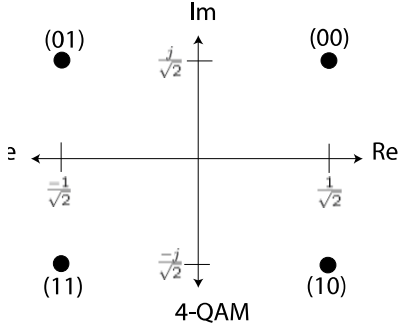
Common constellations (3)

- M-QAM, generalization of QPSK to arbitrary M, power of 4

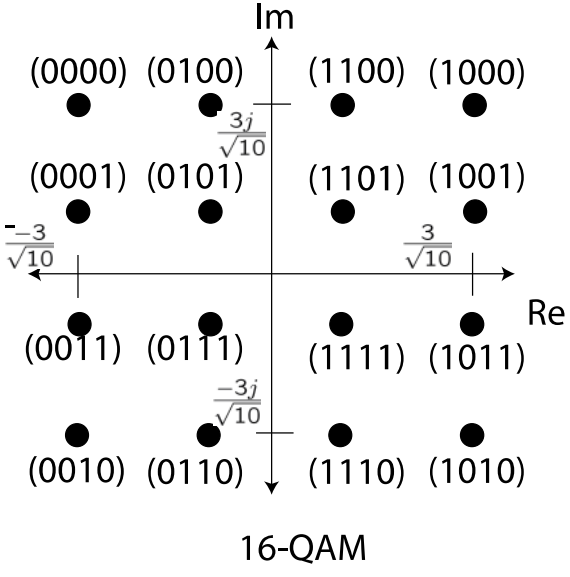
(not normalized)

$$\mathcal{C} = \left\{ \cdots, \frac{k}{2} + \mathrm{j}\frac{\ell}{2}, \cdots \right\} \qquad k, \ell \in \{-(M/2-1), -(M/2-2), \ldots, -1, 1, \ldots, (M/2-1)\}$$

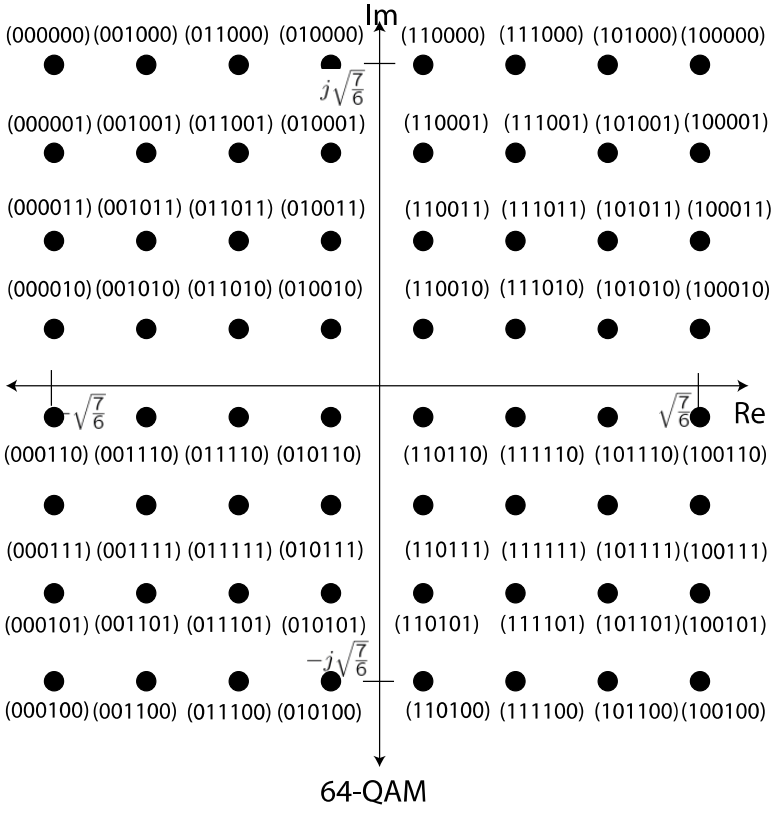
  - 4, 16, and 64 QAM are common in cellular and WLAN
  - 256 QAM is used in 802.11ac
  - Higher order QAM is used in microwave backhaul



Common constellations (4)



16-QAM



64-QAM

Mean of a constellation

$$\mu_s = \mathbb{E}_s\left[s[n]\right]$$

- Because the symbols are equally likely the mean is

$$\mu_s = \frac{1}{M}\sum_{m=0}^{M-1} c_m$$

- Most wireless constellations are zero-mean
  - Otherwise there would be a DC component
  - DC component is problematic at baseband

- To make a constellation zero mean, form a new constellation

$$\mathcal{C}_0 = \{c_0 - \mu_s, c_1 - \mu_s, \ldots, c_{M-1} - \mu_s\}$$

Symbol energy

- The symbol energy is defined as

$$E_{\mathrm{s}} = \mathbb{E}_s \left[|s[n]|^2 \right]$$

- Because the constellation is equally likely and zero-mean

$$E_{\mathrm{s}} = \mathbb{E}_s \left[|s[n]|^2 \right] = \frac{1}{M} \sum_{m=0}^{M-1} |c_m|^2$$

- We always normalize constellation energy to control transmit power

Normalizing a zero-mean constellation

- Given an unnormalized zero-mean constellation

$$\mathcal{C} = \{c_0, c_1, \ldots, c_{M-1}\}$$

- Form the scaled constellation

$$\mathcal{C}_\alpha = \{\alpha c_0, \alpha c_1, \ldots, \alpha c_{M-1}\}$$

- Where

$$\alpha = \sqrt{\frac{1}{\frac{1}{M}\sum_{m=0}^{M-1} |c_m|^2}}$$

Example

- Perform the operations to obtain a zero mean normalized constellation from

$$\mathcal{C} = \{\mathrm{j}, (1-\mathrm{j})/\sqrt{2}, -\mathrm{j}, (-1-\mathrm{j})/\sqrt{2}\}$$

Example – finding the zero-mean constellation

- Compute the mean

$$\mu_s = \frac{1}{4}\left(\mathrm{j} - \mathrm{j} + \frac{1}{\sqrt{2}}(-1-\mathrm{j}) + \frac{1}{\sqrt{2}}(1-\mathrm{j})\right)$$
$$= -\mathrm{j}\frac{1}{2\sqrt{2}}.$$

- Subtract the mean

$$\mathcal{C}_0 = \left\{\mathrm{j}\left(1+\frac{1}{2\sqrt{2}}\right), (1-\mathrm{j})/\sqrt{2}, (-1+\frac{1}{2\sqrt{2}})\mathrm{j}, (-1-\mathrm{j})/\sqrt{2}\right\}$$

Example – finding the normalized constellation

- Compute the scaling factor

$$\alpha = \sqrt{\frac{4}{\left(1+\frac{1}{2\sqrt{2}}\right)^2+1+\left(1-\frac{1}{2\sqrt{2}}\right)^2+1}} = \frac{4}{\sqrt{17}}.$$

- Scaling the constellation

$$\mathcal{C}_0 = \left\{ \frac{4}{\sqrt{17}}\mathrm{j}\left(1+\frac{1}{2\sqrt{2}}\right), \frac{4}{\sqrt{34}}(1-\mathrm{j}), \frac{4}{\sqrt{17}}\left(-1+\frac{1}{2\sqrt{2}}\right)\mathrm{j}, \frac{4}{\sqrt{34}}(-1-\mathrm{j}) \right\}$$

Constellations

Learning objectives:

- Draw a constellation diagram including bit labels
- Calculate the energy of a constellation and normalize it